Slicing

Agenda

- NumPy Array Slicing
  - Slicing 1D NumPy Arrays
  - Slicing Using Step Value
  - Slicing: 2D Array
  - Slicing: 3D Array
  - Negative Slicing

1. NumPy Array Slicing

- In Python, slicing refers to moving elements from one index to another.
- Instead of using an index, the slice is passed as **[start:end]**.
- Another way to pass the slice is to add a step as **[start:end:step]**.
- In slicing, if the starting is not passed, it is considered as 0. If the step is not passed as 1 and if the end is not passed, it is considered as the length of the array in that dimension.

1.1 Slicing 1D NumPy Arrays

Example 1: Illustrate the use of slicing to retrieve employee ratings for a team of seven employees in the first quarter from an array

```
import numpy as np
Employee_rating = np.array([1, 4, 3, 5, 6, 8, 9, 10, 12])
print(Employee_rating[1:7])
```

```
[4 3 5 6 8 9]
```

Example 2: Print the list of three subjects from the fourth index to the end

```
Books = np.array(['Physics','DataScience','Maths','Python','Hadoop', 'OPPs',
print(Books[5:])
```

```
['OPPs' 'Java' 'Cloud']
```

1.2 Slicing Using Step Value

```
X = np.array([8, 7, 6, 5, 4, 3, 2, 1])
print(X[1:6:3])
```

```
[7 4]
```

1.3 Slicing: 2D Array

```
Z = np.array([[11, 22, 33, 44, 55], [66, 77, 88, 99, 110]])
print(Z[0, 2:3])
```

```
[33]
```

1.4 Slicing: 3D Array

```
array_3d = np.array([[[1,2,3],[4,5,6]],[[7,8,9],[10,11,12]]])
print(array_3d[0,1:,1:])
```

```
[[5 6]]
```

1.5 Negative Slicing

- Negative slicing is the same as negative indexing, which is interpreted as counting from the end of the array.

image.png

```
# Negative slicing for 1D array
Neg_slice = np.array([13, 34, 58, 69, 44, 56, 37,24])
print(Neg_slice[:-1])
```

```
[13 34 58 69 44 56 37]
```